



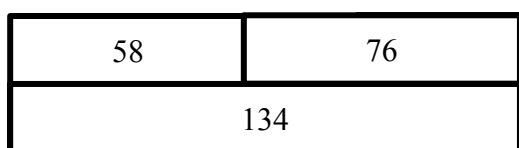
Additive relationships

Task A: Rearranging additive relationships

1) Write the family of rearrangements for the below equations. Draw a bar model to support if needed.

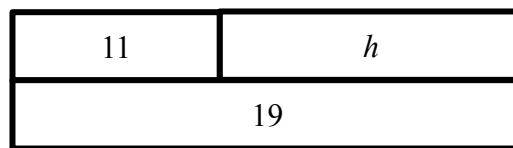
a)

$58 + 76 = 134$	$134 - 76 = 58$
$76 + 58 = 134$	$134 - 58 = 76$



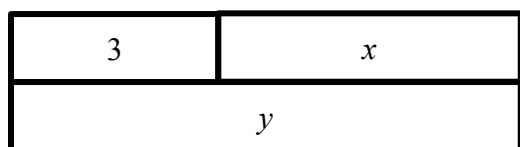
b)

$11 + h = 19$	$19 - h = 11$
$h + 11 = 19$	$19 - 11 = h$



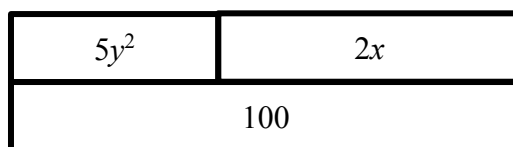
c)

$x + 3 = y$	$y - 3 = x$
$3 + x = y$	$y - x = 3$



d)

$5y^2 + 2x = 100$	$100 - 2x = 5y^2$
$2x + 5y^2 = 100$	$100 - 5y^2 = 2x$



2) Lucas is rearranging equations.



I see a lot of it as 'swapping positions' in the equation.

In each case decide if Lucas is correct and write a sentence to justify your decision.

a)

If $3v + 5w = 52$ then
 $5w + 3v = 52$ is also true.

$3v + 5w = 52$ True. This is an example of
 $5w + 3v = 52$ the commutativity of addition.

b)

If $73 - 8x = 49$ then
 $73 - 49 = 8x$ is also true.

$73 - 8x = 49$ True. $8x$ and 49 can swap positions
 $73 - 49 = 8x$ because we can think of 73 as the sum.

c)

If $3x - 2 = 8y$ then
 $8y - 2 = 3x$ is also true.

$3x - 2 = 8y$ Not true. $3x$ and $8y$ can not swap
 $8y - 2 = 3x$ positions.

$3x$ is the sum so only the 2 and $8y$ can swap positions.



Task B: The sum of multiple terms as a term

1) Fill in the blanks in these rearrangements:

a) $j + k + 8 = 13$

$$j = 13 - (k + 8)$$

b) $j + k + 8 = 13$

$$k = 13 - (j + 8)$$

c) $7x + 8y + 9z = 10$

$$7x = 10 - (8y + 9z)$$

d) $7x + 8y + 9z = 10$

$$9z = 10 - (7x + 8y)$$

e) $3x^2 + 7y^3 - 4 = 10$

$$3x^2 = 10 - (7y^3 - 4)$$

f) $3x^2 - 7y^3 + 4 = 10$

$$3x^2 = 10 - (4 - 7y^3)$$

2) Alex has done really well in this lesson but has made a few errors. Mark his work on rearranging and write a sentence to explain any errors you spot.



a) $x + 3y + z = 7$

$$x = 7 - (3y + z)$$



b) $9x^3 - y^2 + 1 = 8$

$$9x^3 = 8 - (y^2 + 1)$$



$-y^2$ and 1 need to be grouped before subtracting.

$$9x^3 = 8 - (1 - y^2)$$



c) $x + 3y + 5 = 7$

$$x = 7 - (3y + 5)$$

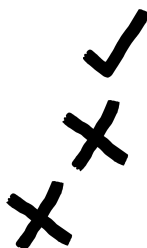
$$x = 7 - 3y + 5$$

$$x = 12 - 3y$$

$$x = 7 - (3y + 5)$$

$$x = 7 - 3y - 5$$

$$x = 2 - 3y$$



We need to subtract both terms inside the brackets.

d) $5d + 2e^3 - 4 = 10$

$$5d = 10 - (2e^3 - 4)$$

$$5d = 10 - 2e^3 + 4$$

$$5d = 14 - 2e^3$$

